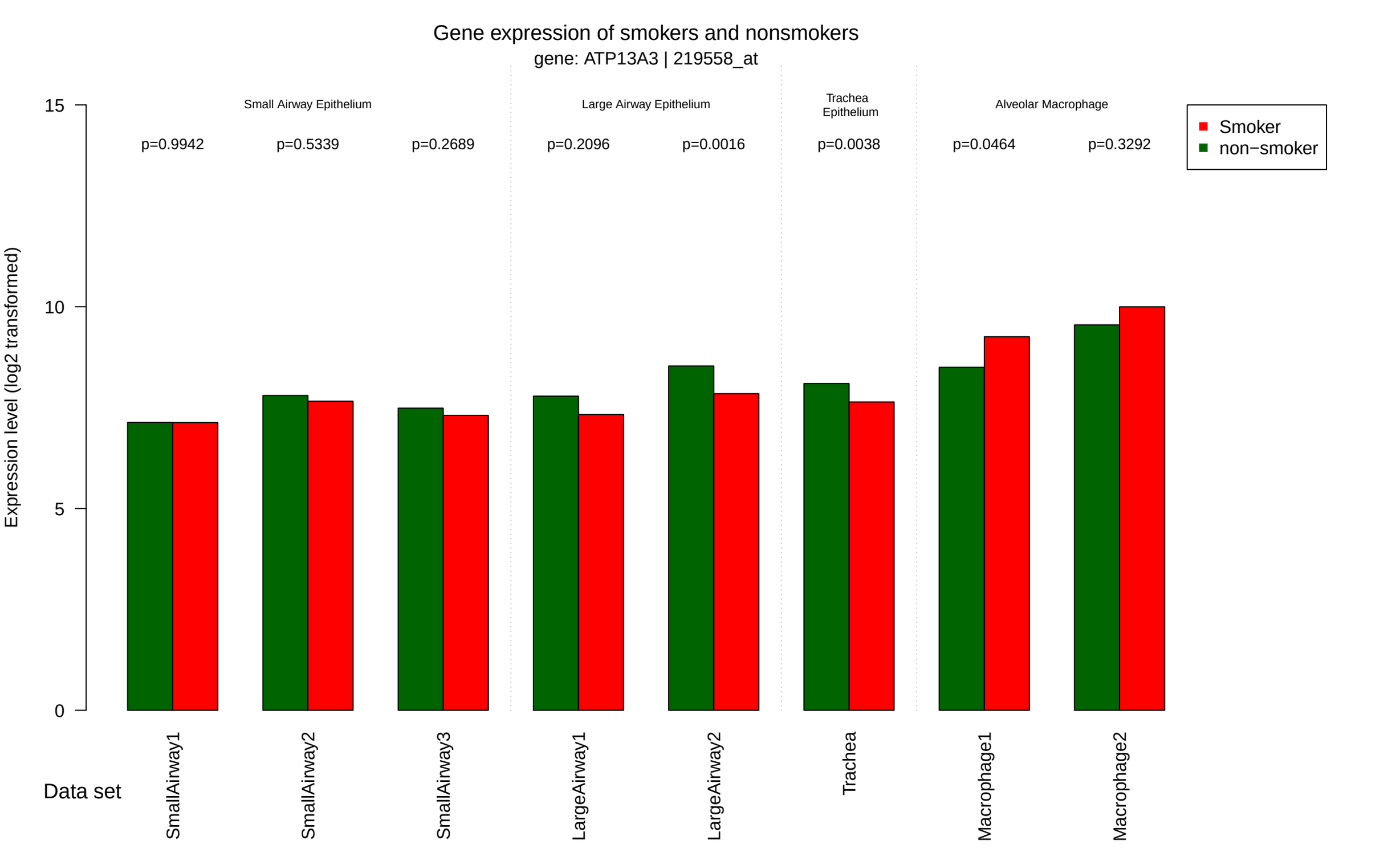
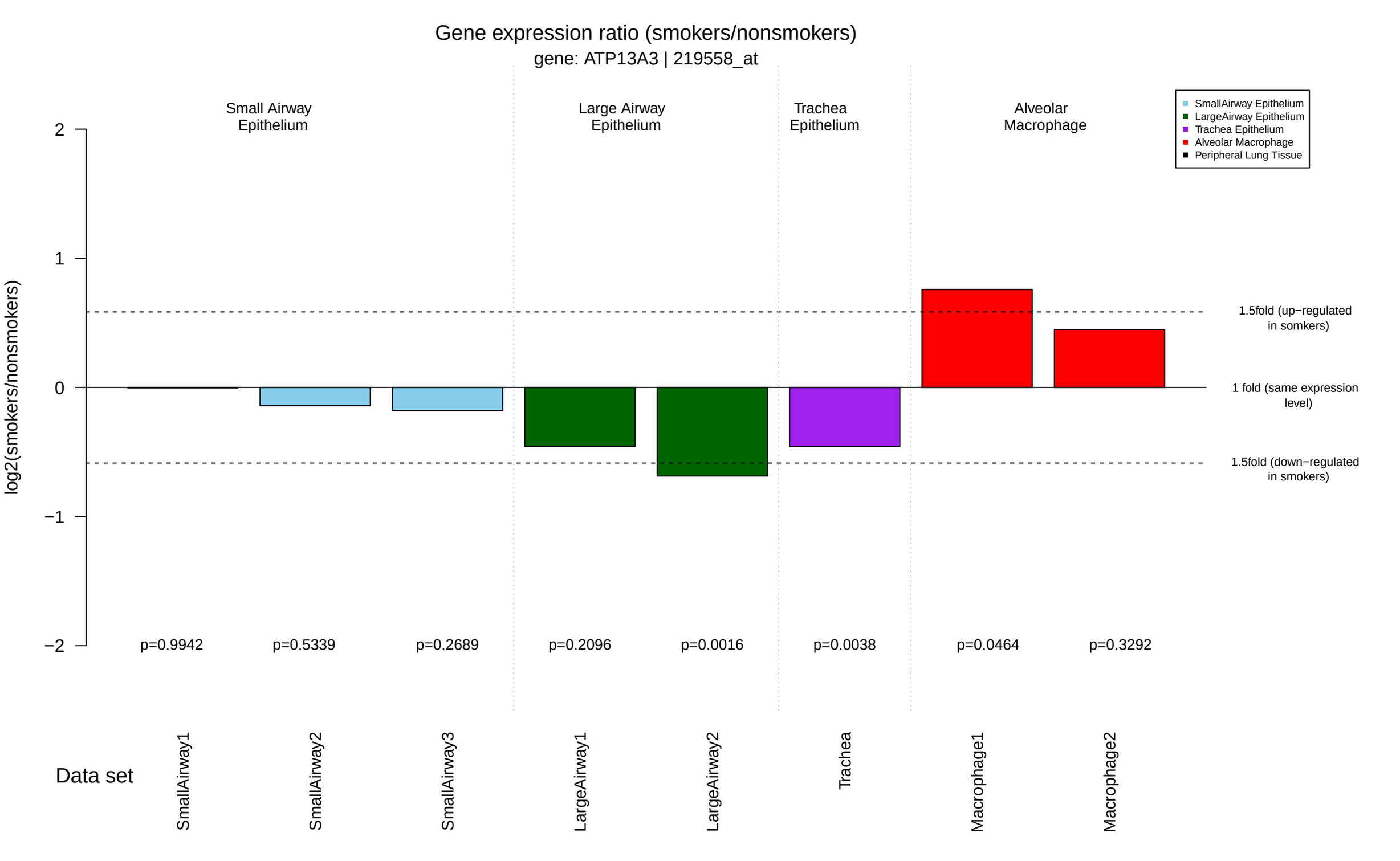
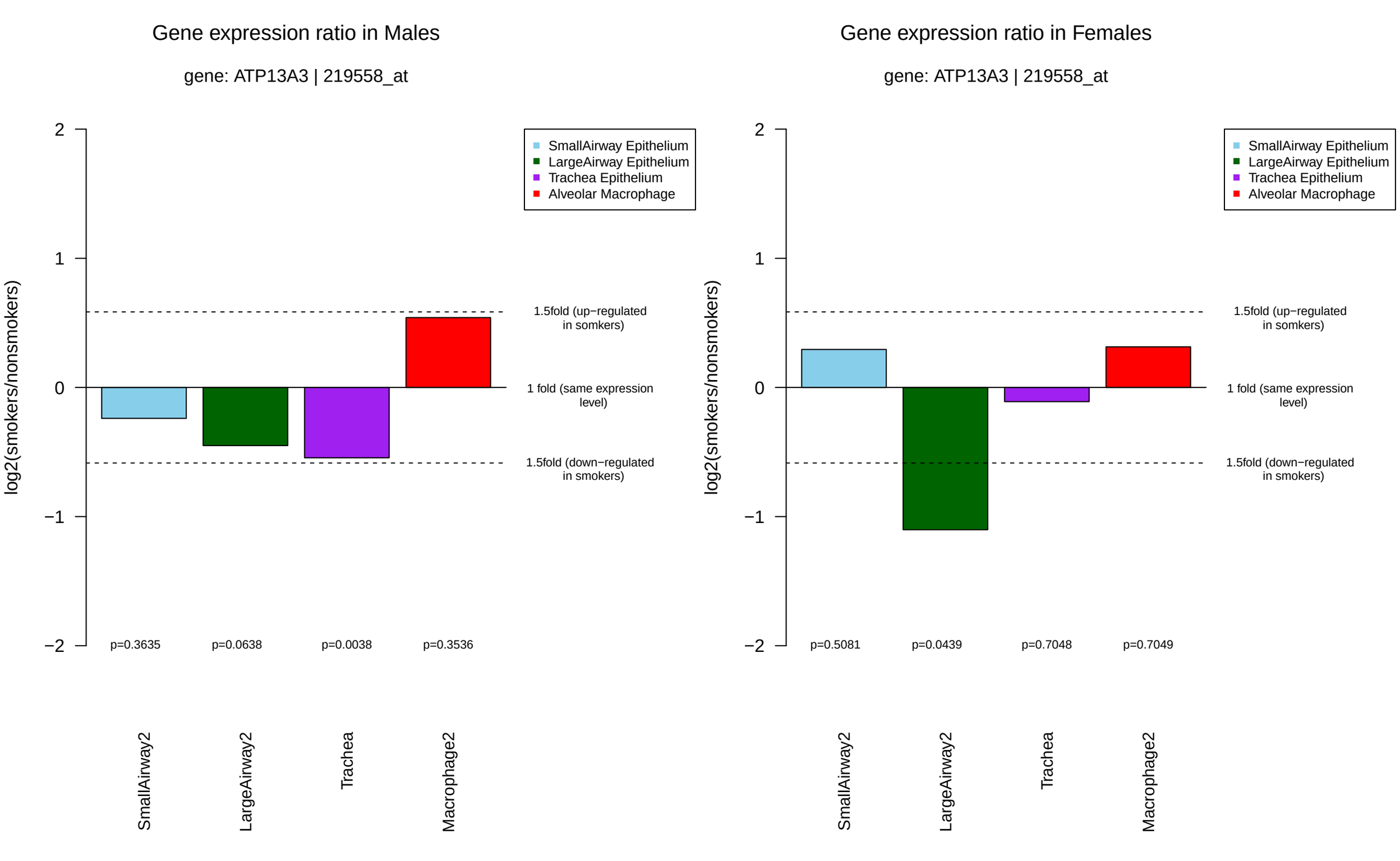
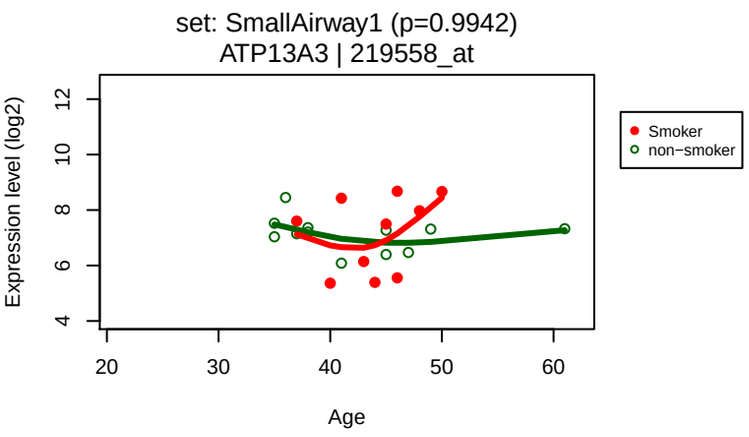
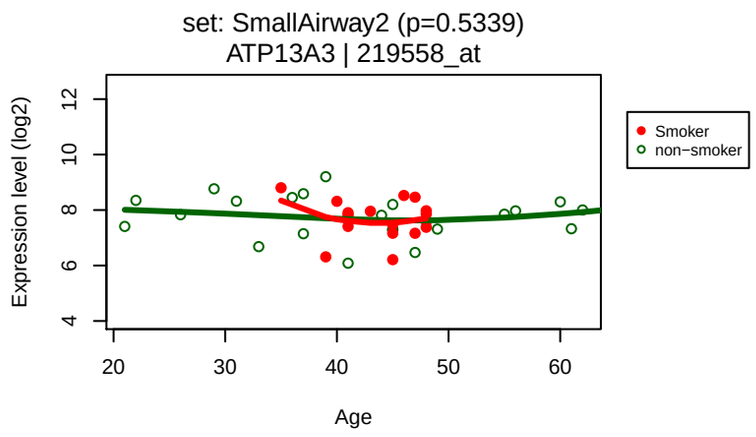
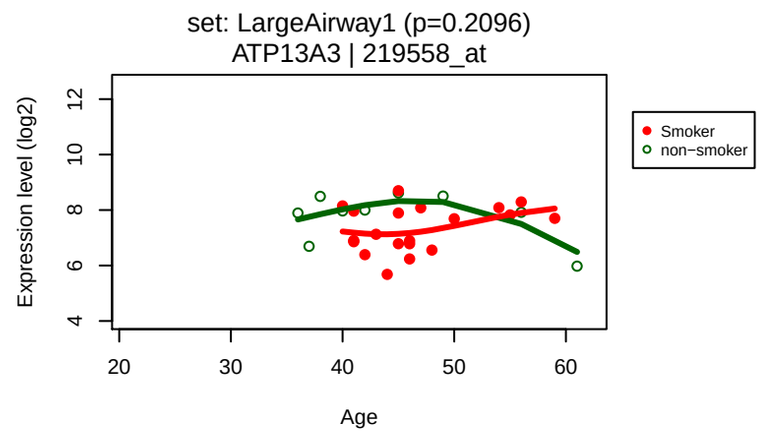

Gene expression of smokers and nonsmokers
gene: ATP13A3 | 219558_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.9942
p=0.5339
p=0.2689
p=0.2096
p=0.0016
p=0.0038
p=0.0464
p=0.3292
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

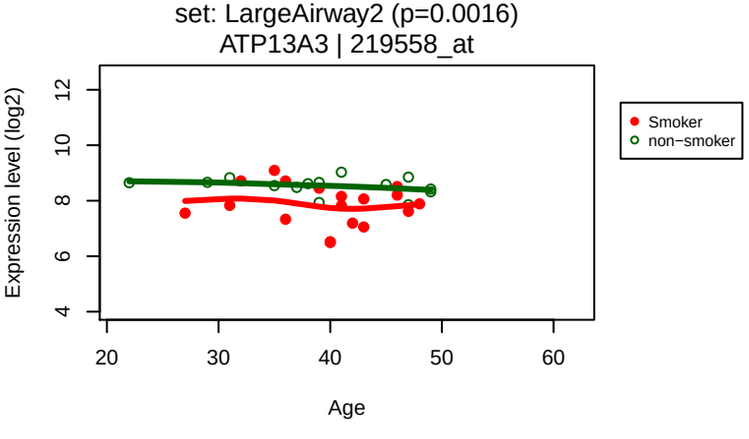

Gene expression ratio (smokers/nonsmokers)
gene: ATP13A3 | 219558_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9942
p=0.5339
p=0.2689
p=0.2096
p=0.0016
p=0.0038
p=0.0464
p=0.3292
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

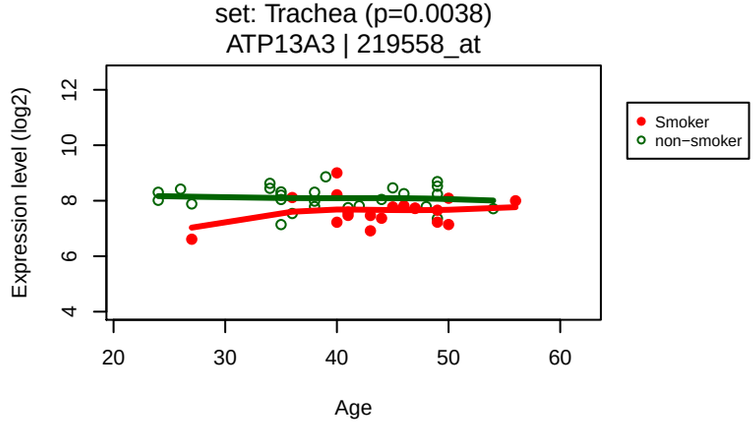

Gene expression ratio in Males
gene: ATP13A3 | 219558_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3635
p=0.0638
p=0.0038
p=0.3536
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ATP13A3 | 219558_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5081
p=0.0439
p=0.7048
p=0.7049
SmallAirway2
LargeAirway2
Trachea
Macrophage2

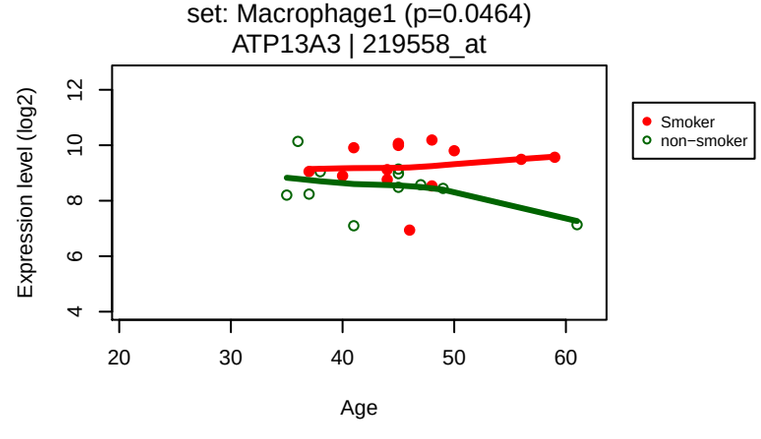

set: SmallAirway1 (p=0.9942)
ATP13A3 | 219558_at
Expression level (log2)
Age
Smoker
non−smoker

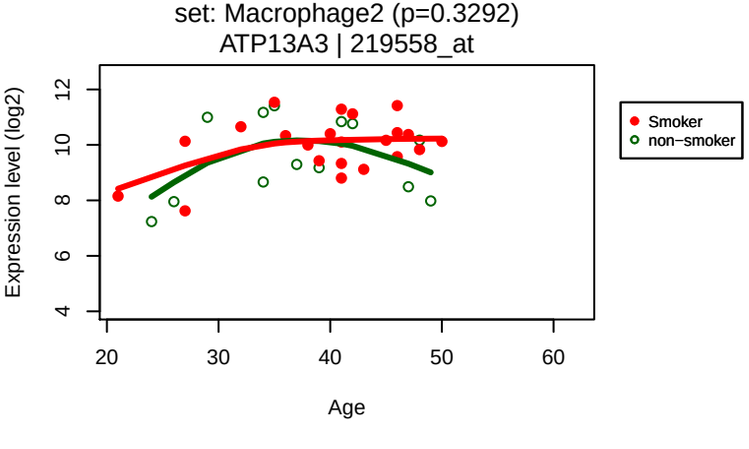

set: SmallAirway2 (p=0.5339)
ATP13A3 | 219558_at
Expression level (log2)
Age
Smoker
non−smoker

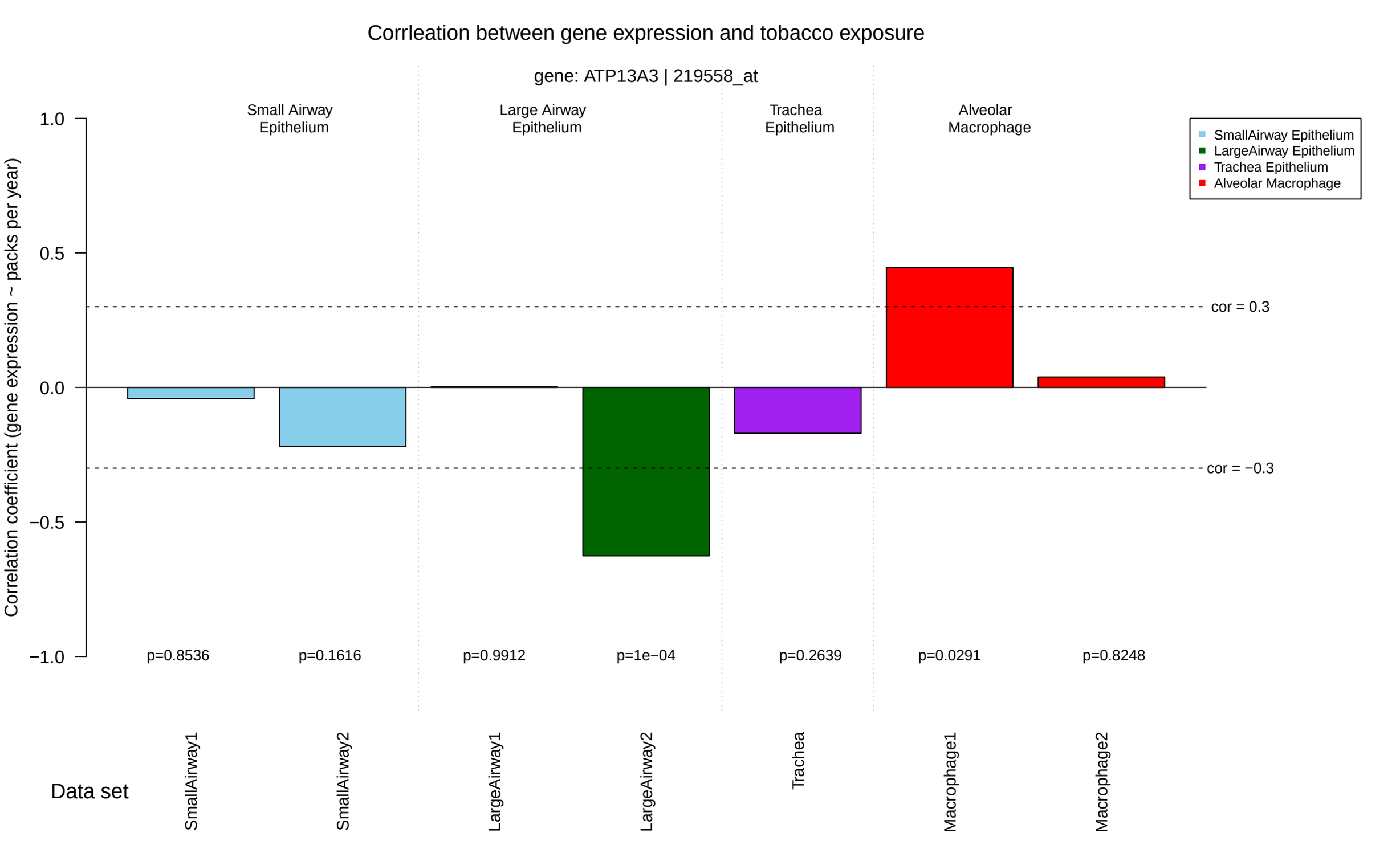

set: LargeAirway1 (p=0.2096)
ATP13A3 | 219558_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0016)
ATP13A3 | 219558_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0038)
ATP13A3 | 219558_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0464)
ATP13A3 | 219558_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.3292)
ATP13A3 | 219558_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ATP13A3 | 219558_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.8536
p=0.1616
p=0.9912
p=1e−04
p=0.2639
p=0.0291
p=0.8248
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2